

A Post-Processing Adapter for Correcting False Positives and False Negatives in Egocentric Hand Detection

Temporal consistency, not detector retraining, as the mechanism for turning a noisy per-frame detector into a trustworthy multi-frame signal

Detection Quality Adapter Project

Egocentric hand-detection post-processing pipeline · August 2026

Abstract—Object detectors operating on video make two kinds of mistakes that are invisible in a single frame: reporting a hand that is not present (a false positive) and missing a hand that is (a false negative). Both are detectable only across multiple frames, since a real hand follows a continuous, physically plausible trajectory that noise does not. This paper describes a detection quality adapter: a stateful, six-stage post-processing layer inserted between a hand detector and its downstream consumers, which corrects both error types using per-frame geometry and multi-frame tracking, without retraining or otherwise modifying the detector. The adapter was built and calibrated against 39 real egocentric work clips (85,786 frames, 153,876 raw detections) spanning 39 distinct jobs, for which no labeled ground truth exists. In the absence of labels, every threshold was checked against the dataset's own real distributions rather than set by inspection, and results are reported as rejection-reason frequencies rather than as precision or recall. The final calibrated pipeline retains 90.5% of tracked detections, recovers 2.6% through interpolation, and rejects the remainder for cause. A depth-based rule for separating a wearer's hands from a bystander's was found not to generalize across recording batches and is documented as opt-in rather than shipped by default.

Index Terms—hand detection, multi-object tracking, false positive rejection, temporal consistency, egocentric video, stereo depth, post-processing pipeline.

Implementation: Python, OpenCV, NumPy. Test suite: 160 passing, 1 deliberate xfail. This paper synthesizes the project's specification, planning log, and final documentation; it cites no external sources.

I. INTRODUCTION

A hand detector scoring video frame by frame fails in two directions. A *false positive* is a detection reported where no hand is present: a duplicate box, a bystander's hand, a shadow mistaken for skin. A *false negative* is a hand present in the frame that the detector fails to report, typically during motion blur or a momentary bad frame. False positives retain footage that should be discarded; false negatives discard footage that was sound.

Neither error is visible from a single frame. A duplicate box looks like a normal detection in isolation. A missed hand looks exactly like an empty frame. The only signal that separates a real detection from a spurious one, or a genuine gap from a real absence, is behavior across many frames: a real hand moves along a continuous, physically plausible path, and noise does not.

The straightforward response — retrain the detector — does not fix the structural problem. A detector scoring one frame at a time has no way to know that a duplicate should lose to the real hand, because within that single frame it cannot tell them apart. This work takes the opposite approach: let the detector over-report (every plausible box, uncapped, unduplicated), and place a stateful, multi-frame adapter after it that has context a single frame never will. This reframes the problem as one of tracking and consistency, not computer vision. Nothing in the adapter performs object detection; every stage asks, given noisy boxes over time, what is the most plausible explanation.

The adapter is specified in two parts. Part 1 defines a generic, class-agnostic version: four stages of geometric and temporal correction that require nothing more than per-class size, shape, and speed expectations. Part 2 specializes it for hands with a low, constantly-reached instance cap, a head-mounted camera's near-continuous motion, a characteristic downward exit path, and a second camera view that makes per-box depth available. This paper covers both parts as built, the real dataset used to calibrate them, and what could and could not be validated in the absence of labeled ground truth.

II. PROBLEM FORMULATION

The adapter sits between a detector and everything downstream, and its contract is deliberately narrow, as summarized in Table I.

TABLE I
THE ADAPTER'S CONTRACT, AS SPECIFIED

Aspect	Specification
Consumes	Per frame, a candidate pool of boxes with confidences, plus frame timestamps
Returns	Corrected detections, each marked reported, merged, rejected, or interpolated
Configured by	Per-class expectations: plausible size, plausible shape, candidate pool size and class maximum, maximum speed, and typical exit behavior
Does not	Alter the detector or require retraining. It does ask the detector to stop capping instances itself, and takes over that selection

Nothing is ever deleted. A rejected box stays in the record, tagged untrustworthy rather than removed, for two reasons: every stage's decision remains auditable after the fact, and a later stage can revise an earlier tag — interpolation, for instance, can overwrite a rejected box's position when the surrounding trajectory supports it. An interpolated detection is an assertion the adapter made that the detector never did, and downstream consumers must be able to tell the two apart.

Six classes of false positive are separable without a model, split by whether they need tracking, listed in Table II.

TABLE II
THE SIX SEPARABLE FALSE-POSITIVE CLASSES AND THEIR ORDER OF APPLICATION

#	Class	Basis	Rejection Criterion
1	Duplicate detections	Geometric	Two heavily overlapping boxes on one object — merge, retain the stronger
2	Implausible size	Geometric	Far larger or smaller than the object can be at working distance
3	Implausible shape	Geometric	Markedly more elongated than the object should be
4	Implausible displacement	Temporal	The box jumps further between consecutive frames than the object can travel
5	Unsupported detection	Temporal	Appears for one or two frames with no track before or after it
6	Static detection	Temporal	Does not move while the camera does, indicating scene structure

The split fixes the pipeline's order: the last three rules need tracks, and tracks only exist after association, so geometric rejection must run before association, which must run before temporal rejection, which must run before interpolation. This ordering is also why the instance cap is deferred rather than applied at the detector. A detector asked to cap itself selects by confidence within a single frame — exactly the comparison that is unreliable, since a duplicate or background object can outscore a real detection in a given frame, discarding the real one before the adapter ever sees it. Instead the detector emits a slightly wider candidate pool, and the adapter selects after association, ranking

candidates by the track supporting them rather than by single-frame score. This admits more false positives at the input, which is acceptable because the temporal rules exist to remove them — but the pool stays narrow, since a wide one only gives association more spurious candidates to confuse.

III. SYSTEM ARCHITECTURE

Detections flow through six stages, always in this order (Fig. 1). Stages 5 and 6 are hand-specific additions layered on top of the four-stage generic core.

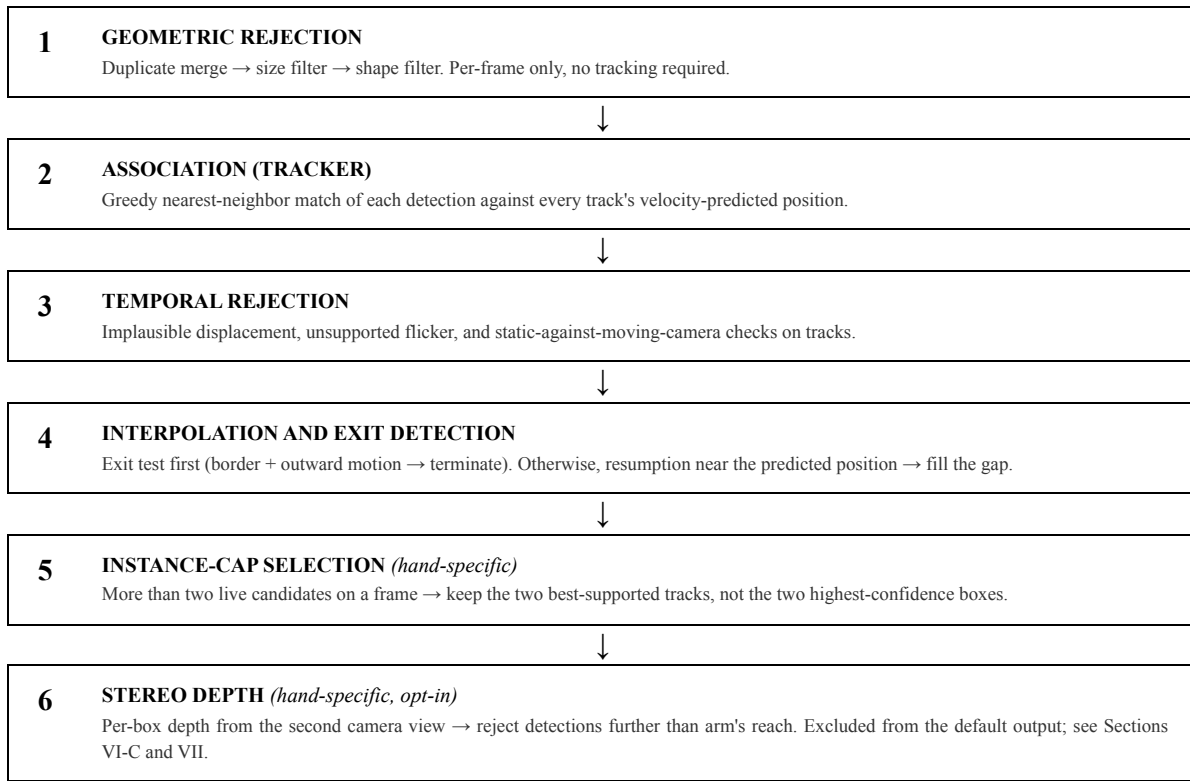


Fig. 1. The six-stage adapter pipeline. Order is load-bearing: stages 3–6 all require tracks, which do not exist until stage 2 has run.

Every detection carries a tag — reported, merged, rejected, or interpolated — that survives to the final output, making every stage's decision inspectable after the fact.

IV. KEY DESIGN DECISIONS

A. Tracking Algorithm: Greedy Nearest-Neighbor, Not Kalman or Hungarian

The tracker predicts each track's next position by linear extrapolation from its last known velocity, then matches the nearest detection within a distance gate. This is deliberately the simplest approach that could work; a Kalman filter (for a smoothed velocity estimate under sensor noise) and Hungarian matching (for globally optimal assignment instead of greedy) were flagged as upgrade paths in code comments but not built, because the simple version passed every synthetic and real-data test. Premature sophistication is a cost, not a feature — the upgrade is built when a real failure demands it, not speculatively.

B. Identity Signal: Position and Motion Only, Never the Detector's Left/Right Label

The detector guesses left- or right-hand per box, but that label is least reliable exactly when it matters most: when two hands cross. The tracker ignores it entirely for identity decisions, verified with an adversarial test in which two

hands cross with their labels swapped at the crossing point. An unreliable signal is worse than no signal at all if the system does not know it is unreliable.

C. Two Gate Radii, Not One

Early versions used one shared `max_speed` value to govern both the tracker's match gate and Stage 3's displacement-plausibility check. Real data showed this to be wrong: a genuinely fast hand movement needs a generous gate so the tracker does not fragment the track, but that same movement should still be flagged as visually untrustworthy once it is part of a track. Conflating "should these be considered the same object" with "is this specific frame trustworthy" is a common tracking bug — they are different questions on different timescales. The fields were split into `track_gate_speed_px_per_frame` (350, generous) and `max_speed_px_per_frame` (110, tightened); see Section VII for the effect this had on real tracks.

D. A Generic Core Plus a Thin, Config-Only Hand Layer

Stages 1–4 know nothing about hands specifically; they are written for any tracked object with configurable size, shape, and speed expectations. A `hand_config()` function supplies hand-specific numbers, and two additive stages (selection, stereo depth) sit on top, without any change to stages 1–4. This is what keeps the generic core testable in isolation and reusable for a different object class.

E. Calibration Philosophy: Real-Data Checks Over Guesses, Labels Over Both

Every placeholder threshold was checked against the real dataset's own distributions before being trusted, using percentiles and structural invariants rather than visual inspection. A threshold that was a complete no-op or actively wrong was fixed immediately, without waiting for labeled ground truth. Anything that genuinely needed labels — true precision and recall, candidate-pool-width tuning — was left alone and documented as blocked, rather than guessed at. The absence of labels is not the same as an absence of anything checkable.

V. DATASET

39 clips, each a distinct real work task, recorded on a head-mounted stereo camera at 1920×1200, 30 fps, both eyes. Every clip bundle carries five artifacts: the stereo

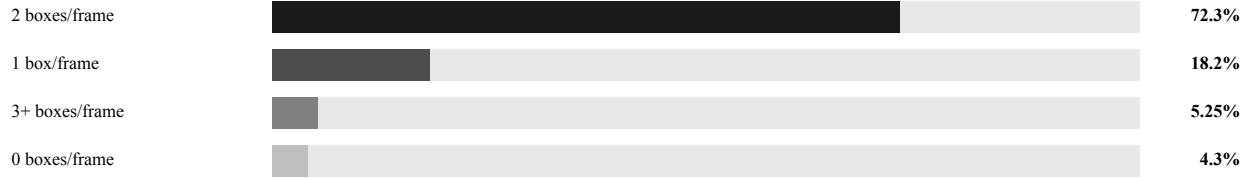


Fig. 2. Detections per frame, across all 39 clips. Frames with three or more boxes (bystanders, unduplicated boxes) are the target of Stage 1.

Remark: every one of the 39 clips is a different job — hairstylist, carpenter, farmworker, jeweler, mason, mechanic, and others, no repeats. This is why several "one global number" thresholds turned out to be too simple: a farmworker's typical working distance (median 0.77 m) and a hairstylist's (1.5–1.8 m, arms extended toward a seated client) are genuinely different populations, not noise around one true value. This recurs throughout Sections VI–VII.

video pair, raw hand boxes from a fine-tuned detector (uncapped, unduplicated, unsmoothed — deliberately noisy), full-rate 6-DoF camera pose from onboard visual-inertial odometry, per-frame timestamps, and task metadata.

TABLE III
DATASET COMPOSITION

Quantity	Value
Clips	39
Total frames	85,786
Total raw detections	153,876
Ground-truth labels available	None

VI. HAND SPECIALIZATION

Hands are one configuration of the generic adapter. Most of the specialization is parameters rather than new logic, but four properties change behavior materially (Table IV).

TABLE IV
HAND-SPECIFIC PROPERTIES AND THEIR CONSEQUENCES, AS SPECIFIED

Property	Consequence for the Adapter
At most two instances per frame	The cap is low and constantly reached, so deferring it matters most here — a duplicate does not merely add an error, it displaces the subject's other hand. The detector requests 3–4 candidates; the adapter selects two after association.
Head-mounted camera	The camera moves constantly, so the static-detection rule is unusually discriminating — a hand is never stationary in the image for a sustained period, in the specification's own assumption (see Section VI-B for where this assumption breaks).
Characteristic exit path	Hands most often leave downward, out of view below the camera, rather than sideways. The leaving-the-frame test weights the lower border accordingly.
A second eye is available	The wearer's hands are within arm's reach and a bystander's are not, so per-box stereo depth separates them directly — the strongest discriminator available, with no equivalent in the generic adapter.

Two further properties are easy to assume incorrectly. Hands are roughly equant, so the shape rule is informative here where it would not be for an elongated object. And handedness is unreliable exactly when hands cross, so tracking associates on position and motion, never on the reported label (Section IV-B).

A. Duplicate Detection vs. Crossing Hands

The generic dedup rule (Stage 1) merges on IoU alone, before any tracking exists — it has no way to know whether two overlapping boxes are one object detected twice, or two real hands that happen to overlap. A real-data check across all 39 clips found the discriminator: every real duplicate pair had a containment ratio (intersection over the smaller box's area) of at least 0.73 — one strong box mostly nested inside a weaker one — while a synthetic same-size 50%-overlap crossing sits at 0.5. A `duplicate_containment_threshold` of 0.7 was added for hands, verified against a pair chosen deliberately in the gap between the two gates (IoU 0.52, containment 0.68) to confirm the fix discriminates on the right property, not just on low IoU.

B. Where the Static-Detection Assumption Breaks

The specification's assumption is that a head-mounted camera moves constantly enough that a hand is never stationary in the image for a sustained period. One carpentry clip contradicts this directly: a hand braced motionless for several seconds while steadying a workpiece, despite continuous camera motion. Requiring

15 consecutive still-while-moving frames before rejecting a detection (rather than one) resolved brief real pauses, but not this multi-second brace. The underlying limitation is structural: telling a genuine multi-second pause from true background clutter needs a model that separates rotation-induced apparent motion from real parallax, not a longer or shorter pixel-threshold window. This remains open; see Section VIII.

C. Stereo Depth: Validated on One Clip, Not on the Dataset

The "beyond arm's reach" rule (Stage 6) was calibrated from a nominal ZED X stereo model — 2.2 mm lens, 12 cm baseline, 733 px focal length — cross-checked against a real high-confidence box at frame 230 of one clip, where a disparity of roughly 267 px implied a depth of about 0.33 m, consistent with that lens's minimum focus distance for a box that large. The two camera eyes were also found not to be perfectly vertically aligned (roughly 2.4 px mean offset, up to 9 px), handled with a vertical search tolerance rather than a strict horizontal epipolar assumption. Running the same calibration across all 39 clips surfaced a confident false rejection of the wearer's own hands on a different clip, traced to the dataset spanning multiple recording batches with different metadata signatures — evidence of different physical camera rigs with no per-clip calibration to correct for it. Stage 6 is therefore opt-in, invoked only when both video paths are explicitly supplied, and excluded from the pipeline's default and final output.

The full specification edge-case table was implemented as one explicit test per row (Table V).

TABLE V
EDGE-CASE HANDLING FROM THE SPECIFICATION, AND IMPLEMENTATION STATUS

Case	Required Handling	Status
Duplicate boxes on one hand	Merge before selection down to two	Implemented
Hand exits at a side border	Terminate the track, do not interpolate	Implemented
Hand passes below the camera	Terminate via a lower-border-specific override	Implemented
Brief occlusion by an object	Interpolate the gap	Implemented
Hands cross or overlap	Retain both; must not be treated as a duplicate	Implemented (Sec. VI-A)
Hand re-enters after a long absence	Start a new track rather than bridging	Implemented
Motion blur during rapid movement	Interpolate where the trajectory is continuous	Implemented
Another person's hands in view	Reject beyond arm's reach by stereo depth	<i>Opt-in, unreliable (Sec. VI-C)</i>
Gloved or partially occluded hand	Behavior unmeasured; needs labeled examples	<i>Deliberate xfail</i>

VII. CALIBRATION AND RESULTS

The specification's validation section asks for two things: precision and recall computed per stage, since each stage should move a specific direction, and a labels-free standing check on the proportion of interpolated detections. Precision and recall are both defined against ground truth — whether a box was actually a hand — and this dataset has none. The matching machinery to compute them is fully

built and unit-tested against synthetic ground truth, ready the moment a labeled reference set exists, but it produces no real figure today. What follows is what was honestly achievable without labels.

A. Threshold Tuning From Real Distributions

Six thresholds were retuned using percentiles of the real dataset rather than inspection, matching the specification's instruction that dropout length in particular should be set from a real distribution rather than guessed (Table VI).

TABLE VI
THRESHOLDS RETUNED FROM REAL, UNLABELED DATA ACROSS ALL 39 CLIPS

Parameter	Before	After	Evidence
plausible_size	(20, 800) px	(50, 1150) px	Real side length p1 = 94 px, max = 1110 px — the old floor was a no-op, the old ceiling rejected real close-up hands
plausible_shape	(0.3, 3.0)	(0.5, 2.0)	Real aspect ratio p1 = 0.54, p99 = 2.10
max_dropout_frames	10	15	Real gap-length p75 = 13 frames
max_speed (plausibility)	150 px/frame	110 px/frame	Real adjacent-frame speed p99.5 = 97.7
track_gate_speed (matching)	shared with above	350 px/frame	Split out; set above the fastest real jump observed (305.8)
duplicate_containment_threshold	n/a	0.7 (hands)	Real duplicate pairs cluster ≥ 0.73 ; crossing hands sit near 0.5

Four further thresholds — camera_moving_speed_mps, camera_moving_angular_deg_per_frame, static_px_threshold, and duplicate_iou_threshold — were

checked against their real distributions and found already in reasonable, non-arbitrary positions; they were left unchanged, now verified rather than merely assumed.

B. Where Every Detection Ends Up

Across all 39 clips, 153,876 raw detections plus 2,964 Stage-4 fabrications give the breakdown in Fig. 3 by rejection reason — a real, unsupervised proxy for where

each rule fires, not a confirmed false-positive rate.

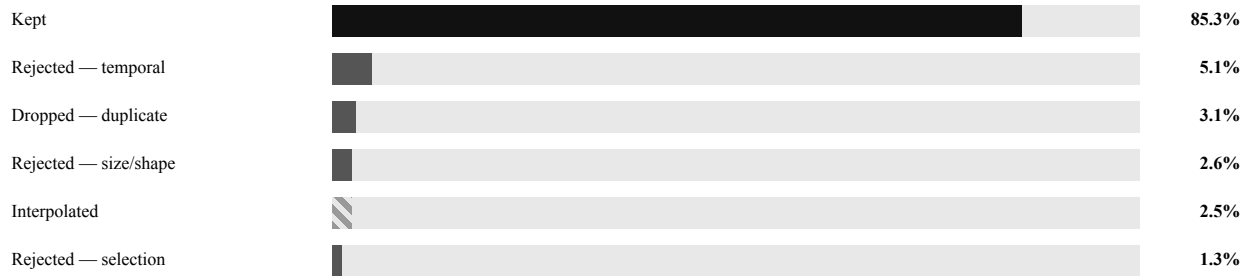


Fig. 3. Detection outcome breakdown across all 39 clips. Black = retained; solid gray = removed for cause; hatched = fabricated by recovery. Percentages are of raw detections plus Stage-4 fabrications.

C. Final Calibrated Pipeline

The tuned hand pipeline (Stages 1–5, Stage 6 excluded) ran against all 39 clips in 996.8 seconds, producing 148,714 final detections across 787 tracks (Table VII).

TABLE VII
FINAL RESULTS ACROSS ALL 39 CLIPS

Metric	Value	What It Stands In For
Final detections (Stages 1–5)	148,714	—
Kept	90.5%	Survived every rule
Rejected — temporal (Stage 3)	5.5%	Proxy for false-positive frequency
Interpolated (Stage 4)	2.6%	Recovered false negatives
Rejected — selection (Stage 5)	1.4%	Cap enforcement, ranked by track quality
Dropped as duplicate (Stage 1)*	3.1%	Proxy for one false-positive class
Rejected — size/shape (Stage 1)*	2.6%	Proxy for another false-positive class
Tracks confirmed exiting	36.1%	Border exits correctly not fabricated
Interpolated proportion, mean/max	3.5% / 17.2%	The specification's labels-free standing check

*Measured against all 153,876 raw detections, not just the 148,714 that survive to a track.

The maximum interpolated proportion (17.2%, one carpentry clip) was checked by hand: its longest interpolated runs are 6–12 frames, consistent with genuine brief-occlusion recovery rather than fabrication. A second full run with Stage 6 enabled (14,319.9 seconds, roughly four hours, no frame cap) reproduced Stages 1–5 exactly, as expected, but reviewing Stage 6 before presenting it surfaced the calibration failure described in Section VI-C.

Given the choice between keeping that run with heavy caveats, spending further time on an unvalidated fix, or reverting to the Stages-1–5 result, the Stages-1–5 output was kept as the authoritative final deliverable; the stereo-depth run is retained separately, labeled as superseded evidence rather than deleted.

VIII. LIMITATIONS

TABLE VIII
SUMMARY OF LIMITATIONS

Limitation	Detail
No ground truth	Every threshold in this paper is evidence-based, not precision/recall-optimal — a genuinely weaker guarantee than the specification's validation section asks for.
Stereo depth does not generalize	Calibrated against and validated on one clip; misjudged depth by roughly 4× on a different clip from a different recording batch. Opt-in, excluded from the default output.
Multi-second static braces	A hand held still for a few frames is correctly protected; one held still for several seconds still reads as background, since the signal cannot yet separate camera rotation from translation.
One global number for 39 jobs	Arm's reach, camera speed, and hand size all vary by task in ways confirmed in the data. A job-aware version was considered and deliberately not built, since deriving a job's "normal" range from its own detections would be circular without labels to check it against.
Greedy, not globally optimal tracking	Nearest-neighbor matching can in principle mis-assign in dense, ambiguous scenes. No real or synthetic test found this happening, but it remains a known theoretical gap.

IX. CONCLUSION AND FUTURE WORK

This paper described a six-stage post-processing adapter that corrects false positives and false negatives in hand detection using temporal consistency rather than detector retraining. Built first as a class-agnostic core validated on synthetic sequences, then specialized for hands with two additional stages, the adapter was calibrated against 39 real, unlabeled egocentric work clips by checking every threshold against real distributions rather than by inspection. The result is a defensible, evidence-based configuration, not a precision/recall-optimal one — a distinction maintained throughout this paper rather than obscured by it.

Three lines of work remain, all requiring the labeled reference set the specification calls for: a corpus weighted toward difficult cases — both hands visible, one hand, no hands, frame-border detections, and borderline scores —

rather than representative footage, since a detector shown no true negatives during evaluation cannot demonstrate it recognizes "none" correctly. Against that reference set, three specific gaps close: true per-stage precision and recall, candidate-pool-width tuning (how often a real hand is outranked by a spurious box), and validation of a job-aware scaling factor for speed and reach thresholds that this paper identified as necessary but could not verify without labels. A fourth line does not require labels but does require new modeling: separating camera-rotation-induced apparent motion from true parallax, to resolve the multi-second static-brace failure of Sections VI-B and VIII. A fifth — recovering or deriving true per-clip stereo calibration, rather than one nominal model applied across recording batches — would let Stage 6 move from opt-in and excluded to a validated part of the default pipeline.

APPENDIX

GLOSSARY OF TERMS

TABLE IX
GLOSSARY OF TERMS

Term	Meaning
Detection	One raw box the detector reported on one frame, with a confidence score
Track	An ordered sequence of detections believed to be the same physical object over time
Tag	What happened to a detection: reported, merged, rejected, or interpolated. Never deleted, always tagged
IoU	Intersection-over-union — how much two boxes overlap, from 0 (none) to 1 (identical)
Disparity	How many pixels the same point shifts between the left and right stereo views — closer objects shift more
VIO pose	Visual-inertial odometry — the camera's own estimated 6-DoF position and orientation per frame, used as the "is the camera moving" signal
Gate/gating	A maximum-distance cutoff deciding whether a detection is even a plausible candidate match for a track
Standing check	A monitor needing no ground truth — here, the proportion of interpolated detections, which rising over time would signal fabrication rather than recovery